Boundary Conditions: Structural Expansion in Dimensional Human Field Theory (DHFT)

DHFT Technical Note: Boundary Conditions

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Abstract

This document defines the structural properties of boundary conditions in Dimensional Human Field Theory (DHFT). The system is specified over invariant variables Load (L), Capacity (C), and Boundary (B).

The relations defined here make explicit the role of boundary conditions in regulating load exposure, containment, and transfer. Directionality and asymmetry are treated as admissible structural properties.

No additional variables are introduced. All relations remain constrained by the minimal system defined in DHFT.

I. Scope

This document defines the structural properties of boundary conditions within DHFT. It specifies how boundary conditions regulate load entry, containment, and discharge.

No interpretive, psychological, or domain-specific constructs are introduced.

II. Variable Reference

The system is defined over three invariant variables: Load (*L*), Capacity (*C*), and Boundary (*B*). These variables are fixed at the minimal layer and are not expanded or redefined within this document.

III. Boundary Function

Boundary conditions regulate system exposure to load. They determine what enters a system, what remains contained within it, and what exits the system.

Boundary conditions do not generate load and do not alter capacity. Their function is regulatory with respect to load movement.

IV. Directionality

Boundary conditions operate directionally with respect to load.

For formal purposes, boundary conditions may be expressed as inflow and outflow relations:

B_in

B_out

Inflow governs load entry into the system. Outflow governs load discharge from the system.

V. Asymmetry

Boundary conditions are not required to be symmetric. Inflow and outflow may differ in magnitude, rate, or constraint.

Asymmetry between inflow and outflow is structurally admissible and may result in net load accumulation or depletion.

VI. Boundary-Regulated Accumulation

Load accumulation is determined by the relation between inflow and outflow under boundary conditions.

Net load is expressed as follows:

$$L_{net} = L_{in}(B_{in}) - L_{out}(B_{out})$$

Accumulation occurs when inflow exceeds outflow. Depletion occurs when outflow exceeds inflow.

VII. Failure Conditions

Boundary failure occurs when boundary conditions do not constrain load in a manner consistent with system stability.

Failure may take the form of:

- excessive inflow
- insufficient outflow
- breakdown of containment

These conditions increase the likelihood of instability or collapse under load.

VIII. Constraint

All valid descriptions of boundary conditions reduce to their role in regulating load exposure and movement. Boundary conditions do not introduce new variables and do not alter the invariant structure of the system.

IX. Statement

Boundary conditions define the structural regulation of load exposure in DHFT. All valid system descriptions at the minimal layer preserve this function.

X. System Reference

The system defined in this document is specified in *Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System*, DOI: 10.5281/zenodo.19075948.

All valid system descriptions reduce to Load, Capacity, and Boundary.

No additional variables are introduced at the minimal layer.

This document is part of the DHFT technical series.

XI. Copyright